

Sharing one heterogeneity parameter across a bridged network cost little; the interval method mattered more

A simulation study answering catalog problem CMP-16

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Abstract

Background. A bridged network contrast pools two subnetworks, and a single between-study SD τ lends one subnetwork’s heterogeneity to the other. Catalog problem CMP-16 asks whether that breaks the bridged contrast’s interval and whether a stratified τ does better at realistic study counts.

Methods. A bridged contrast formed as the sum of two subnetworks’ pooled effects; heterogeneity ratio 1, 2 or 5 in the bridging or the other subnetwork; 3, 6 or 12 studies each; shared, stratified and shrunk REML heterogeneity with Wald intervals, and Hartung-Knapp intervals; 2000 replicates per cell.

Results. With the non-bridging subnetwork five times as heterogeneous, the shared model’s Wald interval covered 0.906, 0.911 and 0.932 at 3, 6 and 12 studies per subnetwork, against 0.917, 0.927 and 0.943 for the stratified model: at most 0.016 apart. Every Wald interval undercovered at few studies, and Hartung-Knapp intervals covered 0.974 to 0.997 at three studies, at 1.5 times the width.

Conclusion. For a contrast that sums both subnetworks, a shared τ overstates one part’s variance and understates the other’s, and the errors largely offset. The undercoverage comes from ignoring uncertainty in τ , not from sharing it.

1 The problem

A shared τ inflates the homogeneous subnetwork’s variance and deflates the heterogeneous one’s. For a contrast that uses only one subnetwork this misstates its interval directly; for a bridged contrast that sums both, the two errors enter with opposite signs. Stratifying τ fixes the structure but estimates two poorly determined variances instead of one.

2 Design

Registered protocol: `protocol.md`. K studies of component A against its backbone (effect -0.3) and K of B against A (0.1); the target is their sum. Within-study SE uniform on 0.1 to 0.25 ; between-study SDs 0.05 in one subnetwork and $0.05R$ in the other. Shared, stratified and shrunk (weight $K/(K+4)$ toward the shared value) REML heterogeneity; Wald intervals, and Hartung-Knapp intervals (1) for shared and stratified.

3 Results

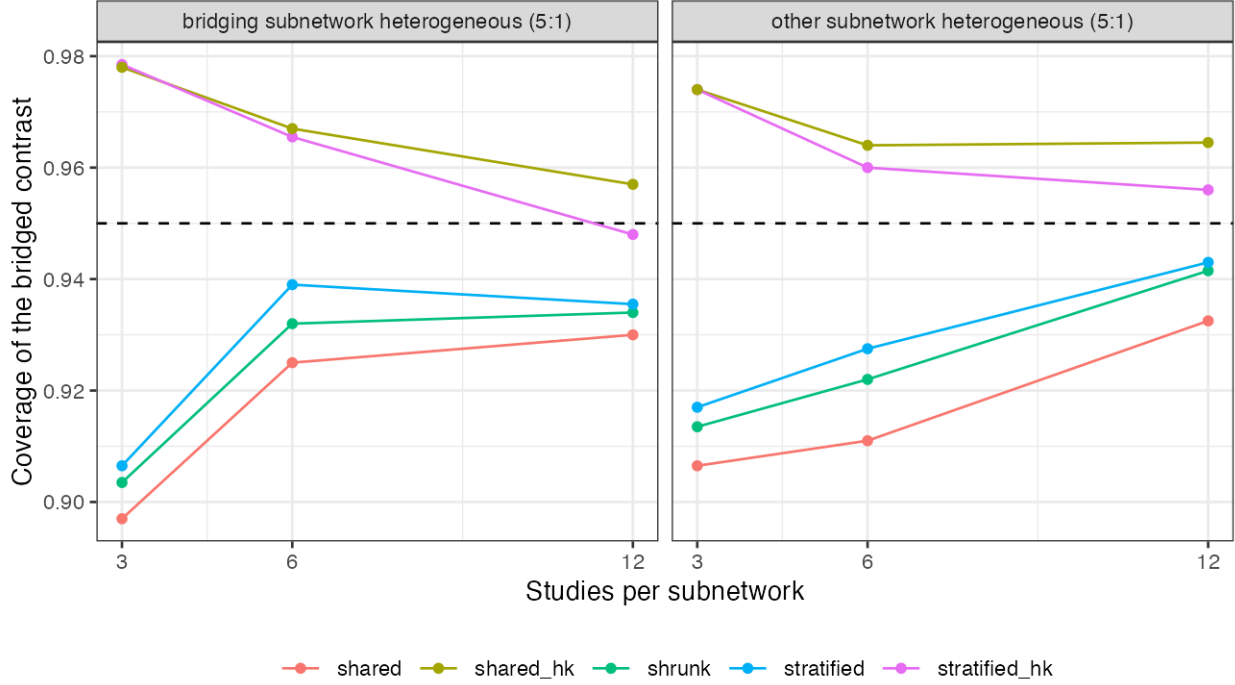


Figure 1: Coverage of the bridged contrast at a 5:1 heterogeneity ratio.

Table 1: Coverage of 95% intervals, 2000 replicates per cell (MCSE at most 0.007).

	ratio	hetero- geneous	studies each	shared	stratified	shrunk	shared, Hartung- Knapp	stratified, Hartung- Knapp
	1	bridging	3	0.955	0.960	0.958	0.997	0.997
	2	bridging	3	0.944	0.951	0.948	0.991	0.992
	5	bridging	3	0.897	0.906	0.903	0.978	0.979
	2	other	3	0.941	0.957	0.948	0.997	0.997
	5	other	3	0.906	0.917	0.913	0.974	0.974
	1	bridging	6	0.951	0.957	0.957	0.979	0.978
	2	bridging	6	0.943	0.950	0.947	0.976	0.973
	5	bridging	6	0.925	0.939	0.932	0.967	0.966
	2	other	6	0.950	0.961	0.958	0.978	0.977
	5	other	6	0.911	0.927	0.922	0.964	0.960
	1	bridging	12	0.954	0.955	0.955	0.968	0.968
	2	bridging	12	0.950	0.951	0.952	0.968	0.967
	5	bridging	12	0.930	0.935	0.934	0.957	0.948
	2	other	12	0.942	0.945	0.947	0.961	0.961
	5	other	12	0.932	0.943	0.942	0.965	0.956

The registered rule returned mixed: in the registered cells (non-bridging subnetwork heterogeneous) the shared model never covered below 0.90, and the shrunk model was not nominal where the shared one fell short (Table 1); the lowest shared coverage, 0.897, was with the bridging subnetwork heterogeneous. Across

the grid the three structures differed by at most 0.016 in coverage and a few percent in width (Figure 1). The gap to nominal was driven by the heterogeneity ratio and the study count, not by the structure: at a 5:1 ratio and three studies per subnetwork every Wald interval covered 0.90 to 0.92, rising to 0.93 to 0.94 at twelve. The shared estimate of τ sat between the two true values, as it must, and the bridged contrast's variance, which sums both subnetworks, was close to right on average.

4 What this does not answer

The bridged contrast here sums both subnetworks; a contrast drawing on one subnetwork only would carry the shared τ 's error without the offset. Frequentist REML without priors; no residual-variance arm, IPD studies or sparse network. Peer review has not been done.

References

1. Joachim Hartung, Guido Knapp. On tests of the overall treatment effect in meta-analysis with normally distributed responses. *Statistics in Medicine*. 2001;20(12):1771–82. doi:[10.1002/sim.791](https://doi.org/10.1002/sim.791)